

IMO(47th) – 2006

First Day

1. Let ABC be a triangle with incentre I . A point P in the interior of the triangle satisfies $\angle PBA + \angle PCA = \angle PBC + \angle PCB$. Show that $AP \geq AI$, and that equality holds if and only if $P = I$.
2. Let P be a regular 2006-gon. A diagonal of P is called good if its endpoints divides the boundary of P into two parts, each composed of an odd numbers of sides of P . The sides of P are also called good. Suppose P has been bisected into triangles by 2003 diagonals, no two of which have a common point in the interior of P . Find the maximum number of isosceles triangles having two good sides that could appear in such a configuration.
3. Determine the least real number M such the following inequality holds for all real numbers a, b, c .

$$\left| ab(a^2 - b^2) + bc(b^2 - c^2) + ca(c^2 - a^2) \right| \leq M(a^2 + b^2 + c^2)^2$$

Second Day

4. Determine all pairs (x, y) of integers such that $1 + 2^x + 2^{2x+1} = y^2$.
5. Let $P(x)$ be a polynomial of degree $n > 1$ with integer coefficients and let k be a positive integer. Consider the polynomial $Q(x) = P(P(\dots(P(P(x))\dots)))$, where P occurs k times. Prove that there are at most n integers t such that $Q(t) = t$.
6. Assign to each side b of a convex polygon P the maximum area of a triangle that has b as a side and is contained in P . Show that the sum of the areas assigned to the sides of P is at least twice the area of P .

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